

Weekly Report

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Project: UAV-Based Traffic Safety & Video Analytics

Week: 4 Apr 2026 - 11 Apr 2026

Submission Date: 11 Apr 2026

Detailed Summary of Work Done During Current Week

A. Vehicle Trajectory Generation Completion

- Successfully generated vehicle trajectories for multiple vehicles across extended video segments.
- Utilized centroid-based tracking from annotated bounding boxes to create continuous movement paths.
- Verified trajectory continuity through visualization on video frames.
- Ensured trajectories are aligned with the registered coordinate system and predefined zones.

B. Trajectory Validation and Visualization

- Visualized vehicle trajectories overlaid on video frames to confirm correctness of movement paths.
- Checked trajectory smoothness and continuity across frames.
- Identified and filtered minor inconsistencies in trajectory paths.
- Confirmed that generated trajectories are suitable for further behavioral analysis.

C. Basic Behavior Analysis (Initial Stage)

Performed initial analysis on selected vehicle trajectories based on predefined criteria:

- **Zone Entry and Exit Detection**
 - Identified when vehicles enter and exit specific predefined zones.
 - Computed basic entry/exit timestamps for selected vehicles.
- **Movement Pattern Observation**
 - Analyzed vehicle paths relative to road geometry.
 - Observed lane-following behavior in selected cases.

- **Preliminary Rule-Based Insights**
 - Identified simple cases of:
 - Early movement toward stop-line
 - Delayed stopping behavior
 - Established initial logic for detecting such patterns using trajectory data.

D. Pipeline Completion (End-to-End Workflow)

- Established a working **end-to-end pipeline**:
 - YOLO inference → CSV → XML → CVAT refinement → trajectory extraction → basic analysis
- Ensured smooth integration between all stages.
- Organized outputs including:
 - Annotation files
 - Trajectory datasets
 - Visualization outputs
- Confirmed that the pipeline is functional for performing traffic behavior analysis on UAV videos.

Tasks Planned for Coming Week

A. Behavior Analysis Expansion

- Extend analysis to more vehicles and longer video segments.
- Implement additional behavior metrics such as:
 - Stop-line compliance
 - Creeping behavior
 - Lane discipline

B. Result Preparation and Documentation

- Generate visual outputs for report and presentation.
- Document methodology and analysis pipeline.
- Prepare structured results for final project submission.